

EatClub Brands (Box8 & Mojo Pizza)

Dynamic Price Prediction Model

AI-Powered Demand Forecasting & Real-Time Dynamic Pricing for India's Cloud Kitchen Unicorn

Predictive ML · Kafka Streaming · Ruby on Rails · AWS · Real-Time Pricing

<3%

Food Waste

25 Min

Delivery Time

150+

Kitchens

\$40M

Tiger Global

EXECUTIVE SUMMARY

What is EatClub's Price Prediction Model?

EatClub Brands (formerly Box8) operates India's largest cloud kitchen network — Box8 & Mojo Pizza — processing millions of orders across 150+ kitchens in 6+ cities.

Their Dynamic Price Prediction Model uses proprietary ML algorithms to:

- Predict hyper-local demand per kitchen per hour
- Dynamically adjust pricing based on real-time supply/demand signals
- Pre-position food production before orders arrive
- Minimize waste while maximizing kitchen throughput
- Automate procurement across 150+ kitchens in real-time

Key Business Outcomes



Food Waste

<3% vs 15% industry avg



Delivery Speed

~25 min consistent delivery



Kitchen Network

150+ kitchens, 6+ cities



Funding

\$40M led by Tiger Global



Unit Economics

Positive — rare in cloud kitchen sector

PROBLEM STATEMENT

The Cloud Kitchen Paradox — High Volume vs. High Waste

01

Unpredictable Demand Spikes

Lunch & dinner rushes cause massive order surges that legacy systems couldn't anticipate — leading to either stockouts or over-preparation and 15%+ food waste industry-wide.

02

Static Pricing Losing Revenue

Fixed menu prices meant EatClub couldn't capture willingness-to-pay during peak hours or stimulate demand during off-peak — leaving significant revenue on the table daily.

03

Siloed Kitchen Inventory

150+ kitchens operated in isolation with no real-time stock visibility. Manual procurement caused over-ordering at one location and stockouts at another simultaneously.

04

Slow Order-to-Delivery Cycle

Without pre-prediction, every order started cooking from scratch — making consistent 25-min delivery impossible and degrading customer experience at peak hours.

SOLUTION OVERVIEW

Full-Stack AI: From Farm-to-Fork Demand Prediction to Real-Time Dynamic Pricing

Robust & Scalable Architecture

- Proprietary ML algorithms predicting demand at hyper-local (per kitchen) and temporal (per hour) granularity
- Dynamic pricing engine adjusts menu prices in real-time based on demand signals, weather, time-of-day, and competitor data
- Kafka-based event streaming pipeline for real-time order, inventory & pricing data across 150+ kitchens
- Vertically integrated supply chain — EatClub owns procurement, production & delivery, unlike aggregator-dependent rivals
- Microservices architecture: Cart, Coupons, Logistics, Kitchen Display System — all decoupled & independently scalable

Demand Forecasting

Pre-positions base gravies, dough & prep during off-peak — final assembly only at order time

Dynamic Pricing Engine

Real-time price adjustments — surge pricing at peak, discounts at off-peak to maximize revenue

Inventory AI

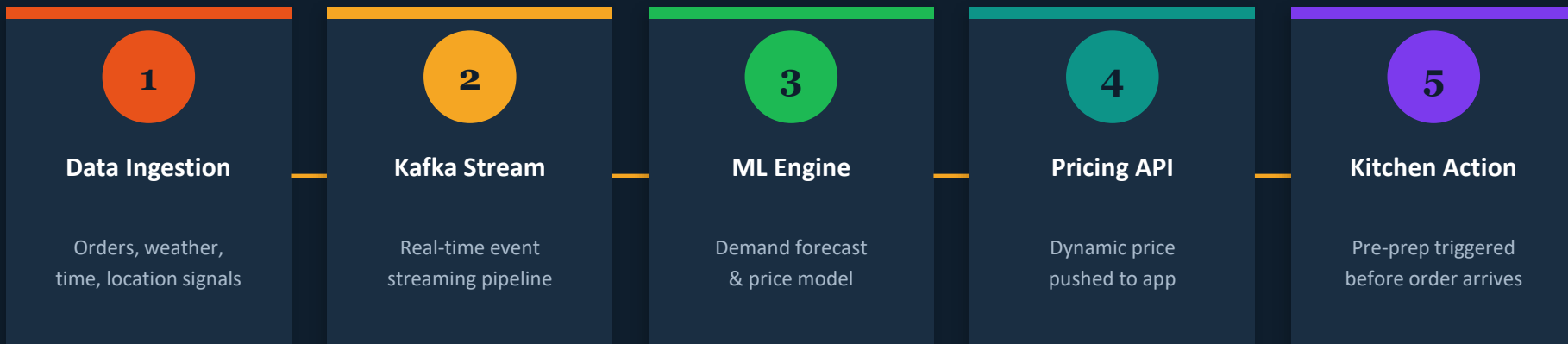
Cross-kitchen real-time stock monitoring with auto-procurement triggers to prevent stockouts

Route Optimization

ML-driven delivery routing adapts dynamically to traffic, kitchen load & delivery partner availability

TECHNICAL ARCHITECTURE

End-to-End Data Pipeline — From Order Signal to Price Decision



Ruby on Rails + C++

Primary backend (RoR) with C++ for performance-critical modules — JSON parsing optimized for 300k+ daily users.

Kafka + Redis + PostgreSQL

Kafka for event streaming, Redis for high-speed caching (50% latency reduction), PostgreSQL as primary DB.

AWS Cloud + Redshift

Elastic AWS compute handles peak lunch/dinner spikes; Redshift for analytics and ML model training data.

Microservices (Cart/Coupon/KDS)

Decoupled services: Cart, Coupons, Logistics, Kitchen Display System — each independently scalable.

DYNAMIC PRICING MODEL — DEEP DIVE

Input Layer: Multi-Variable Signal Ingestion

- Historical order data: per kitchen, per SKU, per hour — last 90 days rolling window
- Real-time signals: weather API, day-of-week, local events, festival calendar
- Competitor pricing feeds: scraped from Swiggy/Zomato listings in real-time
- Kitchen capacity state: current prep load, staff levels, equipment availability

Forecast Layer: Hyper-Local Demand Prediction

- ML model outputs demand probability per SKU per kitchen per 30-min slot
- Gradient Boosting + LSTM ensemble — temporal patterns + non-linear demand signals
- Accuracy target: <8% MAPE (Mean Absolute Percentage Error) on hourly forecasts
- Pre-positions base gravies, dough & ingredient prep 45–60 min before predicted peak

Pricing Layer: Real-Time Price Optimization

- Price elasticity model estimates demand sensitivity per item per market segment
- Surge pricing: up to 15–20% price increase during high-demand slots
- Discount triggers: automated coupon generation during off-peak to fill kitchen capacity
- Margin floor enforced: pricing engine never goes below cost + minimum margin threshold

Execution Layer: Kitchen & Delivery Orchestration

- Kitchen Display System (KDS) receives pre-prep instructions before orders arrive
- Inventory procurement auto-triggered when stock falls below predicted demand buffer
- Delivery route optimization recalculates every 90 seconds based on live traffic data
- Feedback loop: actual vs. predicted demand updates model weights in near real-time

ML MODELS & AI TOOLS — INTERNAL DETAILS

Under the hood: What powers EatClub's pricing intelligence



Demand Forecasting Models

- Primary: Gradient Boosting (XGBoost/LightGBM) for tabular demand features
- Secondary: LSTM (Long Short-Term Memory) neural network for time-series order patterns
- Ensemble: Both models combined — GBM for cross-sectional signals, LSTM for temporal patterns
- Features: Hour, day, kitchen_id, SKU, weather, local_event, trailing_7d_avg, promo_flag
- Retraining: Daily model refresh on rolling 90-day window of order history
- Accuracy: <8% MAPE on 30-minute demand slots per kitchen per SKU



Dynamic Pricing Engine

- Price Elasticity Model: Estimates demand sensitivity (dQ/dP) per item per location
- Optimization Algorithm: Constrained optimization — maximizes revenue subject to margin floor & demand volume targets
- Surge Logic: If `predicted_demand > 85% kitchen capacity` → `price += elasticity_factor × surge_multiplier`
- Discount Logic: If `predicted_demand < 40% capacity` → `coupon_value = (capacity_cost / expected_orders) × margin_buffer`
- Competitor Scraping: Swiggy/Zomato price feeds scraped every 15 min via headless browser
- Output: Real-time price pushed to app via FastAPI endpoint — <200ms latency SLA

DATA SOURCES, STACK & SPECIFICATIONS

Data Inputs



Order Transaction Data

Per-SKU, per-kitchen, per-hour historical orders — 90-day rolling window



Weather & External APIs

Real-time weather, local events, festival calendar — key demand drivers



Competitor Price Feeds

Swiggy/Zomato listings scraped every 15 min — headless browser automation



Kitchen Operations Data

Staff levels, equipment state, prep capacity, current order queue load

Core Tech Stack

Ruby on Rails

Primary backend web framework for all API and business logic layers

C++ Modules

Performance-critical modules: JSON parsing, real-time pricing computation

Apache Kafka

Real-time event streaming across all 150+ kitchen data pipelines

Redis Cache

High-speed caching — reduced API response time by 50% on Coupon service

PostgreSQL

Primary relational database — orders, inventory, pricing history

BUSINESS VALUE ACHIEVED

<3%

Food Waste

vs 15% Industry Avg

25min

Delivery Time

Consistent at Peak

50%

Latency Cut

Coupon API Response

\$40M

Tiger Global

Series Funding

Operational Efficiency

- Pre-positioning production based on ML forecasts eliminated last-minute rush cooking — consistent 25-min delivery regardless of peak hour
- Freed kitchen staff from reactive cooking to proactive, scheduled preparation — improving quality and consistency across brands

Cost & Waste Reduction

- Food waste cut from industry avg 15% to under 3% — directly improving gross margins in a sector where margins are notoriously thin
- Auto-procurement AI eliminated over-ordering and stockouts simultaneously across 150+ geographically dispersed kitchens

Revenue Optimization

- Dynamic surge pricing captured peak-hour willingness-to-pay; off-peak discounts filled unused kitchen capacity — maximizing daily revenue per kitchen
- Positive unit economics achieved — rare milestone in Indian cloud kitchen sector, enabling Tiger Global's \$40M investment

Scalability

- Microservices architecture scaled Coupon API to 300k+ daily users with 50% latency reduction via Redis caching
- Entire ML + pricing stack operates identically whether EatClub runs 10 kitchens or 150 — linear scaling without linear cost increase

STRATEGIC CONCLUSION

EatClub — Solving the Cloud Kitchen Paradox with Predictive AI

EatClub's Dynamic Price Prediction Model is not merely a pricing tool — it is the operational brain of the entire cloud kitchen network. By predicting demand before it happens, the system pre-positions production, eliminates waste, captures revenue at every demand level, and achieves consistency at scale that human operations simply cannot. It solves the fundamental tension of cloud kitchens: high volume without high waste.

Full-Stack Advantage

Owning farm-to-fork means every ML insight translates directly into action — no aggregator dependency

Predictive > Reactive

Cooking before orders arrive is the core innovation — ML turns a reactive kitchen into a predictive factory

Scalable Unit Economics

The same ML stack that works at 10 kitchens works at 150 — making geographic expansion capital-efficient

Key Facts Recap

Models: XGBoost + LSTM Ensemble · Stack: Ruby on Rails + C++ + Kafka + Redis + PostgreSQL · Cloud: AWS · Waste: <3% · Delivery: 25 min · Kitchens: 150+ · Funding: \$40M Tiger Global · Pricing Latency: <200ms